NeuronBench and `evolve_pysr`

Meta-evolving symbolic regression for mechanistic neuron dynamics

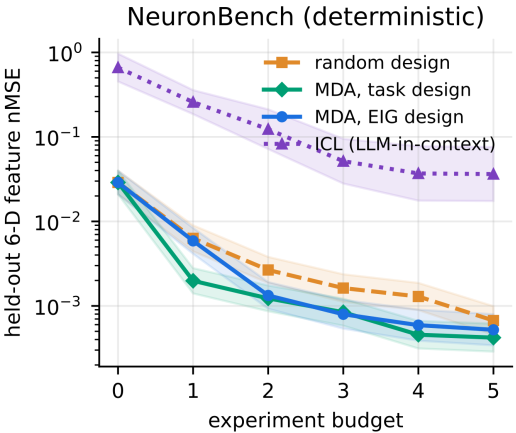
- Adapt six generalized Hodgkin–Huxley “mystery neurons” into exact, fully observable symbolic-regression problems.
- Evolve PySR mutation, survival, selection, and loss operators on subsets of neuron worlds, then evaluate transfer to held-out worlds.
- Newest “uninformative prompt” run (708907): **23/25 strict numerical recoveries**, plus two near-exact fits and no misses.

Primary references: Murphy, *Model Discovery Agent*, arXiv:2608.09696v4; project runs 313196, 313195, 190177, and 708907.

Original NeuronBench: active model discovery

Murphy's benchmark defines six "mystery neurons" using generalized Hodgkin–Huxley ODEs. Each has a novel membrane mechanism intended to be difficult to identify from textbook recall alone.

- The agent chooses among nine current-clamp protocols.
- It observes voltage trajectories; gating state and mechanism are latent.
- It predicts six trajectory features under held-out interventions.
- MDA combines nested SMC^3, LLM model proposals, and Bayesian experiment design.

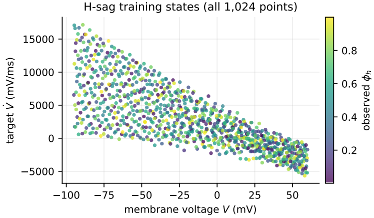


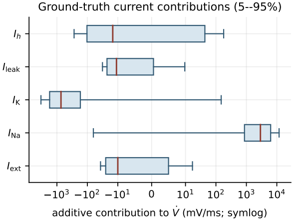
Murphy Fig. 4(a): deterministic NeuronBench; mean ± SE.

Primary source: Kevin Murphy, *Model Discovery Agent: LLM-assisted Bayesian experiment design for data-efficient discovery of mechanistic world models*, arXiv:2608.09696v4, Sec. 4.4 and Fig. 4(a).

Our reduction: expose the state and recover the current balance

$$\dot{V} = I_{\mathrm{ext}} - \sum_c g_c\,\phi_c\,(V - E_c) - g_L(V - E_L).$$





What is changed

- No active experiment selection.
- Observe I_{ext}, V, and each composite open fraction ϕ_c.
- Fit only the scalar vector field $\dot{V}$.

Data protocol

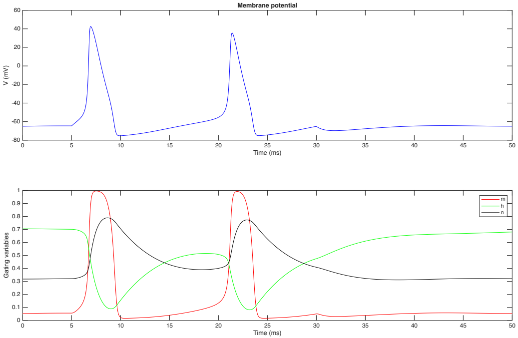
- 1,024 scrambled-Sobol training states.
- 16,384 held-out states.
- $V \in [-95, 60]$ mV; $\phi_c \in [0, 1]$.
- Analytic, noiseless targets.

Primary project data: `runs/neuronbench_fully_observable/data/h_sag.npz`; generator and equation: `scripts/neuronbench_fully_observable.py`.

Connection to the existing Sandia Hodgkin–Huxley project

$$C_m \dot{V} = I_{\mathrm{ext}} - g_{\mathrm{Na}} m^3 h (V - E_{\mathrm{Na}}) - g_{\mathrm{K}} n^4 (V - E_{\mathrm{K}}) - g_L (V - E_L).$$

The Sandia demonstration uses a canonical HH trajectory; NeuronBench retains the current law but tests six mechanisms on independent collocation states.



Existing Sandia simulation: membrane potential and the m, h, n gating variables.

Primary local source: `/home/sca63/sandia_spiking/Hodgkins-Huxley/hh_fig1.pdf`; equations: `hh.tex`; simulation: `hh.py`.

The six symbolic-regression problems

Every world contains the common Na, K, and leak terms. The table gives the world-specific change.

World	Observed state	Additional term in $\dot{V}$	Phenotype in original benchmark
Z-rebound	$\phi_Z = m_Z^2 h_Z$	$-4\phi_Z(V-120)$	low-threshold inward current; block
H-sag	$\phi_h = m_h$	$-5\phi_h(V+30)$	sag and post-inhibitory rebound
Na-fatigue	$\phi_{\mathrm{Na}} = m_{\mathrm{Na}}^3 h_{\mathrm{Na}} s_{\mathrm{Na}}$	none; slow state enters ϕ_{Na}	use-dependent spike-count rundown
Ca-rebound	$\phi_T = m_T^2 h_T$	$-3.2\phi_T(V-120)$	rebound burst
D-type	$\phi_D = m_D h_D$	$-9\phi_D(V+77)$	delayed/suppressed firing
Textbook-M	$\phi_M = m_M$	$-2.5\phi_M(V+77)$	spike-frequency adaptation

$$f_*(I, V, \phi) = I_{\mathrm{ext}} - 120\phi_{\mathrm{Na}}(V-50) - 36\phi_{\mathrm{K}}(V+77) - 0.3(V+54.4) + \text{world-specific term.}$$

Primary source: pinned `neuronbench.worlds` implementation at commit `c354622458c460b419cab821d482c879f0578377`; project manifest: `runs/neuronbench_fully_observable/data/manifest.json`.

What `evolve_pysr` actually evolves

Candidate bundles replace four parts of PySR. Fitness is the fraction of training worlds for which any saved Pareto-frontier equation reaches NRMSE $\leq 10^{-6}$.

Regime	Run	Mutation	Survival	Selection	Loss
top-1	313196	residual linear boost	age-regularized	membrane-dynamics tournament	transition-weighted MSE
top-2	313195	residual linear step	age-regularized	frequency-robust tournament	phase-aware MSE
"top5"	190177	residual additive correction	Pareto crowded replacement	Pareto rank + crowding	ordinary MSE

Evolution protocol

- population 10; 10 offspring per generation;
- 15 generations for top-1/top-2, 10 for the earlier top5 fold;
- three PySR seeds per candidate, ten-seed reevaluation for finalists;
- final transfer: five independent seeds per held-out world, 10^6 evaluations per fit.

Repeated result across independent runs

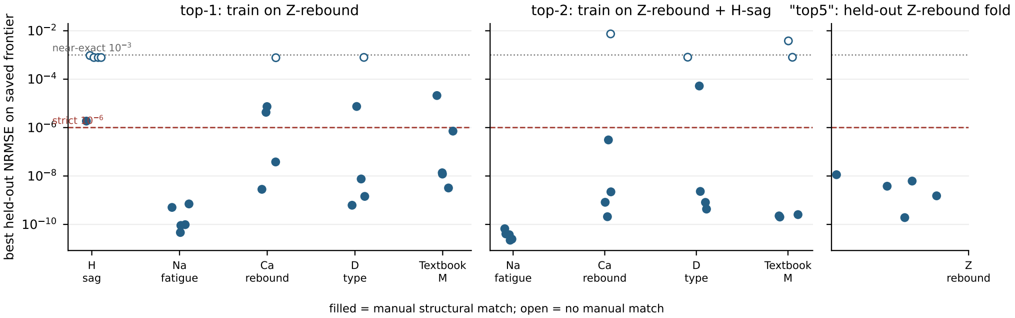
All three best bundles evolved a mutation of the form

$$f_{\mathrm{child}}(x) = f_{\mathrm{parent}}(x) + \widehat{\beta}_j x_j, \qquad \widehat{\beta}_j \approx \arg\min_{\beta} \|r - \beta x_j\|_2^2.$$

This is well matched to an additive ionic-current law.

Primary project artifacts: `run_data.json` and saved `best_bundles` in runs 313196, 313195, and 190177.

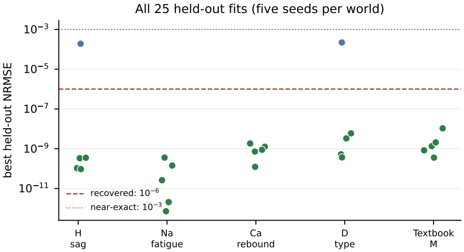
Earlier transfer results: all seed-level NRMSE values

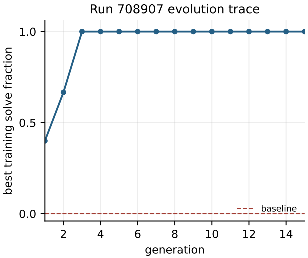


Regime	Training worlds	Held-out worlds	Manual matches	Rate	Interpretation
top-1	1	5	19/25	76%	one evolved outcome
top-2	2	4	16/20	80%	one evolved outcome
"top5"	5	1	5/5	100%	one completed LOOCV fold

Primary results: `runs/313196/neuron_full_eval/neuron_results.json`, `runs/313195/neuron_full_eval/neuron_results.json`, `runs/312780/neuron_results.json`; manual audit: `reports/neuron_manual_match_comparison.json`.

Newest run 708907: "uninformative prompt," raw held-out results





Prompt objective

"Improve the algorithm's ability to discover the expression that generated the task data."

Training

- Z-rebound only;
- baseline: 0/3 training solves;
- best bundle reaches 3/3 by generation 3.

Held-out transfer

- 23/25 recovered ($\leq 10^{-6}$);
- 2/25 near-exact ($\leq 10^{-3}$);
- no close or miss outcomes.

Primary artifacts: `runs/708907/run_data.json`, `runs/708907/best_bundles/best_final.jl`, and `runs/708907/neuron_full_eval/neuron_results.json`.

A recovered equation matches the physical coefficients

Example: "top5" held-out Z-rebound, seed 10000, best frontier row. In physical units,

$$f_*(x) = x_0 - 120x_1x_2 - 36x_1x_3 - 4x_1x_4 - 0.3x_1 \\ + 6000x_2 - 2772x_3 + 480x_4 - 16.32.$$

Monomial	truth	discovered	absolute error
$x_0 = I_{\mathrm{ext}}$	1	1.000000002	2.1×10^{-9}
$V\phi_{\mathrm{Na}}$	-120	-120.000000192	1.9×10^{-7}
ϕ_{Na}	6000	5999.999975	2.5×10^{-5}
$V\phi_{\mathrm{K}}$	-36	-36.000000140	1.4×10^{-7}
ϕ_{K}	-2772	-2772.000019	1.9×10^{-5}
$V\phi_Z$	-4	-3.999999978	2.2×10^{-8}
ϕ_Z	480	479.999995	5.1×10^{-6}
V	-0.3	-0.299999829	1.7×10^{-7}
1	-16.32	-16.319975	2.5×10^{-5}

Numerical and structural checks

- held-out NRMSE $= 1.52 \times 10^{-9}$;
- all nine ground-truth monomials are present;
- no additional monomials;
- maximum absolute coefficient error 2.5×10^{-5}.

Primary coefficient expansion: `reports/neuron_top5_symbolic_recovery_analysis.json`, recovery 0; source frontier: `runs/312780/neuron_results.json`.

Interpretation and remaining evidence gaps

What the current evidence supports

- PySR can recover the complete physical current-balance polynomial, not only a low-error surrogate.
- Bundles evolved on subsets of neuron mechanisms transfer to held-out worlds.
- Residual-guided additive mutation appears independently in the best top-1, top-2, top5, and uninformative-prompt bundles.
- Run 708907 shows that this motif does not require neuron-specific objective wording.

What remains to establish

- Replicate meta-evolution runs before interpreting top-1/top-2/top5 as a learning curve.
- Complete the other five top5 leave-one-world-out folds.
- Apply the same physical-support manual audit to run 708907.
- Test whether residual-projection mutation transfers beyond this additive current-balance domain.

Working conclusion

Meta-evolution repeatedly discovers a residual-fitting mutation that is structurally aligned with additive mechanistic laws.

Interpretation based on the primary run artifacts cited throughout; metric distinctions and audit caveats are documented in `reports/neuron_topk_manual_transfer_report.pdf`.